



USENIX

THE ADVANCED COMPUTING
SYSTEMS ASSOCIATION

Security and Privacy Advice for UPI Users in India

Deepthi Mungara and Harshini Sri Ramulu, *Paderborn University*;
Yasemin Acar, *Paderborn University and The George Washington University*

<https://www.usenix.org/conference/usenixsecurity25/presentation/mungara>

This paper is included in the Proceedings of the
34th USENIX Security Symposium.

August 13–15, 2025 • Seattle, WA, USA

978-1-939133-52-6

Open access to the Proceedings of the
34th USENIX Security Symposium is sponsored by USENIX.

Security and Privacy Advice for UPI Users in India

Deepthi Mungara[§], Harshini Sri Ramulu[§], Yasemin Acar^{§†}
[§]Paderborn University, [†]The George Washington University

Abstract

Unified Payments Interface (UPI) payment systems are widely used in India and are also gaining global traction. UPI enables people to make quick everyday transactions and recurring payments, including rent, gas, and electricity, using the same app. The widespread adoption of UPI has sparked significant concerns regarding users' security and privacy, especially due to an alarming number of UPI-related scams and fraudulent transactions. While prior work has explored the technical security of UPI [1], [2], to address security threats effectively, we must understand user mental models, concerns, security information sources, and behaviors. In a mixed-methods study of 26 semi-structured interviews with UPI users from India and content analysis of 16 security information sources from regulatory bodies, UPI apps, and banks offering UPI, we explore user mental models, concerns, where and how they receive security advice, as well as their security-relevant behaviors. We provide an analysis of users' concerns and threats around UPI security and privacy and highlight gaps where official advice falls short. Further, we recommend UPI providers and banks to curate accessible and useful advice to better alleviate users' concerns, and increase their reach. We also recommend individual security and privacy practices for UPI users to protect themselves.

1 Introduction

India's prevalent online payment system, Unified Payments Interface (UPI), was developed in 2016 by the National Payments Corporation of India (NPCI) [3] and Reserve Bank of India (RBI) [4] and was quickly adopted for cashless transfers; Rastogi et al. describe convenience and efficiency as factors for adoption [5]. UPI enables instant money transfers with bilateral instant confirmation without sharing bank details and personal information, utilizing mobile phone numbers, QR codes, or UPI IDs. UPI can be used with mobile phones through stand-alone apps such as PhonePe [6] and Google Pay [7], and Automated Teller Machines (ATMs), and works

across different banks as listed by NPCI [8].

This rapid adoption of UPI and its increase in digital payments to over 83 million transactions, aggregating to 139 trillion INR (Indian Rupees, 1.67 trillion USD) [9] has also raised an important issue with *the security and privacy of UPI transactions* [10]. In a recent incident in Mumbai, often referred to as the financial capital of India, fraudsters successfully stole 10 million INR (120,000 USD) from 81 UPI users [11]. Their scam involved sending a small amount of money to their target (potentially randomly chosen, by phone number) through UPI applications like Google Pay [7], then contacting their victims, claiming the transfers were made by mistake and requesting the users to send the money back. When victims agree and use a QR code or mobile number to refund money, their bank account and/or personal data may be compromised, they may be phished, led to install malware, or may be made to transfer a larger amount of money than they were initially sent [12], [13].

While UPI was initially designed to revolutionize digital transactions within India, its reach is gradually extending internationally towards UAE, Mauritius, Sri Lanka, Singapore, Bhutan, and Nepal [14] and even towards France in the European Union [15]. At the time of writing this research paper, the European Payments Initiative (EPI) [16] built by European banks and payment service providers was introducing an instant payment solution called Wero and started rolling it out in the European Union [17]. Therefore, understanding how UPI users in India are affected by security threats can help improve secure use of UPI for users in India, and also beyond, possibly even before UPI or similar systems are rolled out more widely.

Even though many cases of fraud attacks target not the technical implementation but instead the users, existing literature primarily focuses on technological aspects of UPI and has often missed the critical knowledge gap in UPI users' mental models and their experiences [18], [19], and how users are advised on secure UPI use. To close this knowledge gap, we answer the overarching research questions:

RQ1: What are UPI users' security perceptions, concerns,

behaviors, and rationales?

RQ2: Where do they learn their security awareness and behavior?

RQ3: How does this correspond to official advice from banks, UPI providers, and regulatory bodies?

To answer the research questions, we conducted a mixed-methods study as follows:

a) Examination of the UPI Advice Landscape: We examined the security and privacy advice and recommendations available to UPI users in India from official sources, namely regulatory bodies, banking institutions, and UPI applications (Table 2). We analyzed advice from the websites of the only two regulatory bodies in India that monitor UPI—*NPCI* and *RBI*, the ten most-used UPI-enabled banks—*State Bank of India (SBI)*, *Industrial Credit and Investment Corporation of India (ICICI)*, *Housing Development Finance Corporation (HDFC)*, *Andhra Pragathi Gramina Vikas Bank (APGVB)*, *Axis Bank*, *Bank of Baroda*, *Union Bank of India*, *Punjab National Bank*, *Kotak Mahindra*, and *Canara Bank* [20], and four most used UPI applications—*PhonePe*, *Google Pay*, *Paytm*, and *BHIM* [1]. We qualitatively analyzed the advice through content analysis, finding six distinct security-relevant categories (Section 5.1).

b) Exploration of User Perspectives: We conducted semi-structured interviews with 26 UPI users in India, collecting data on user mental models, and details of their motivations, concerns, experiences, and challenges when using UPI. We qualitatively analyzed interviews through thematic analysis to understand UPI users' perspectives and experiences (Section 5.2).

In the advice analysis, we categorize available advice from websites to help systematically understand and bridge the gap between users' concerns and threats and the advice they receive, with proper communication strategies by providers (Section 5.1). Our interview study results contribute in-depth insights into the UPI mental models, security concerns, challenges, level of awareness, safety measures followed, and information sources. We also highlight that despite extensive security and privacy advice, many UPI users remain unaware of essential safety measures, and often rely on default settings or follow informal sources for guidance (Section 5.2). We link concerns and mitigations identified in interviews to the advice categories to understand the gap between those. Our findings underline the need for UPI authorities to enhance their communication strategies and ensure that important security advice is both easily accessible and effectively communicated to users.

2 Background

Here, we provide the required details to understand UPI's functionality and the key actors involved.

UPI Ecosystem Actors.

The following are the elements involved in the UPI ecosys-

tem [21]: *Customer Payment Service Player (Customer PSP)* including PhonePe, GPay, and Paytm; *Payee Payment Service Player (Payee PSP)*, including mobile service providers, subscription services, and e-commerce platforms; *Remitter Banks*, i.e., banks that initiate and send payments on behalf of their customers or account holders; *Beneficiary Bank*, i.e., banks that receive funds on behalf of the beneficiary, the intended recipient of a financial transaction; *NPCI*, the regulatory body of UPI; *bank account holders*, and *Merchants*.

How does UPI work?

Any Indian bank holder can generate a UPI ID and initiate transactions with a mobile phone. For example, <user-name>@<bankId> is a unique payment identity, allowing the transfer of money without disclosing users' original names or their bank account information [2]. Figure 1 shows the payment flow of UPI transactions in the PhonePe application [22], one of the most used applications in India for online transactions [23]. PhonePe's UPI transaction process involves a straightforward three-step procedure: First, the merchant presents a Quick Response (QR) code to the customer. Second, the customer scans it with a UPI app (e.g., PhonePe, Google Pay). Third, the customer enters the amount to be transferred and enters their PIN (Personal Identification Number) to complete the transaction to confirm sending the amount to the merchant.



Figure 1: Payment Flow of PhonePe Application [22].

3 Related Work

Researchers have explored the transition of Indian payment systems to UPI, its adoption, challenges, and the revolution it created in the digitalization of transactions. We have categorized the related work from previous studies majorly into the following categories: *Indian Banking and Payment Systems and Transition to UPI*, *User Perceptions on UPI*, *Impact of UPI*, and *UPI Security and Privacy*.

Indian Banking and Payment Systems and Transition to UPI.

To understand how digital payment systems have changed in India after UPI, Mahapatra et al. and Kumar et al. in 2018 and 2019 analyzed the Indian financial system, focusing on the concept of digital banking, which offers services through

the internet and mobile platforms by integrating various technologies [24], [25]. To understand the technical aspects, transaction flow, challenges, and other important aspects of UPI. Prior work has studied the implementation and functionality of UPI following the demonetization period, where the legal status of certain bank notes was removed [21], [26]–[28]. Although our research is about user perception, it is helpful to understand the technical perspectives of UPI.

User Perceptions.

User perceptions were studied to understand UPI acceptance: Khanra et al. investigated consumer resistance factors and found that privacy concerns and usage barriers are crucial factors influencing UPI usage [19]. While analyzing different UPI applications available in the market, Chavan et al. researched the usage patterns, popularity, and reasons for using Google Pay, PhonePe, Paytm, Amazon Pay, and BHIM and underlined the preferences and behaviors of users regarding these payment apps [29]. Sankararaman et al. explored various aspects of UPI transactions, including user awareness levels, user satisfaction, encountered problems, and resolution times [30]. Unlike our research, their analysis is limited to a correlation between user age and satisfaction level and provides regulatory recommendations focusing on enhancing UPI adoption and cybersecurity measures. Another 2022 research conducted by David et al. focuses on user understanding of a specific UPI application called Google Pay, developed by Google [31].

Impact of UPI.

Nagarjuna et al. explored how digital payments are used for quick, secure, and affordable transactions for users [32]. Other research examined the impact of UPI on specific gender and economic demographics [5]. Thakkar et al. investigated the impact of UPI among India's economically poor population and women and highlighted the wide adoption among all categories [33]. Research conducted by Vadlamudi et al. discussed the UPI payment model's adoption by several countries, which we argue requires robust security measures [34]. Exploring a different population, Borborah et al. studied how UPI impacts street vendors in local markets and examined how vendors navigate these obstacles as part of the shift towards a digital economy post-demonetization [35].

UPI Security and Privacy.

After the widespread adoption of UPI, concerns about the security and privacy of digital payment systems have increased. Prior work has also studied the exponential rise of online payments in India facilitated by UPI and focused on addressing the increasing threat of fraudulent activities, particularly social phishing [36], [37]. Lakshmi et al. focused on the security aspects of UPI applications, identifying areas for improvement to safeguard against fraudulent transactions [2]. Their research investigated the architecture, transactions, features, and security issues of UPI apps and proposed enhancements to improve authentication and authorization methods. Relatedly, Kumar et al. conducted a thorough security analysis

of the UPI protocol, which forms a critical backbone for smartphone-based payment apps in India [1]. Their study involved reverse-engineering seven popular UPI apps, discovering previously unknown design flaws in the UPI 1.0 version, particularly in the multi-factor authentication mechanism. These vulnerabilities could potentially lead to severe attacks, including unauthorized access to bank accounts. Furthermore, Madwanna et al. examined the operational mechanism of UPI, highlighting its differences from conventional cashless transactions and exposing security vulnerabilities observed in the UPI app BHIM 1.0 version [38]. While existing research extensively covers the evolution and significance of UPI, its user-friendliness, and its global impact on payment systems, only a limited number of studies looks into user perceptions, preferences, security and privacy suggestions. There is a noticeable gap in the literature regarding the security and privacy advice landscape and recommendations available to UPI users in India, as well as the official guidance on how to protect themselves while using UPI. Given this challenge, in this study, we explored the diverse concerns, experiences, and challenges faced by the general UPI users and the kind of security and privacy advice that are available and being utilized among them.

4 Methodology

In this section, we explain our study's methodology, including data collection and analysis for advice analysis and semi-structured interviews. In total, we analyzed 16 sources of official UPI security and privacy advice and conducted 26 semi-structured interviews with UPI users.

4.1 Study Procedure and Data Collection

We conducted our study in two steps, *a) analysis of the UPI security and privacy advice* from official sources like UPI providers, banks, and regulatory bodies and *b) semi-structured interviews with UPI users* to explore their motivations, challenges, and security and privacy concerns. In this section, we discuss the details of how we collected advice, analyzed the data and conducted interviews.

Advice Data Collection.

We manually collected UPI security and privacy advice from 16 official sources like websites of regulatory bodies, banks, and UPI entities (Table 2). We chose these sources because they are direct providers of UPI applications. The advice they provide on websites is credible and aligned with industry standards supporting the sources within respective fields [39]. NPCI and RBI are the only two official regulatory bodies monitoring the UPI in India, whereas the banks and applications we selected are mostly used by customers according to the studies and surveys [1], [20]. We navigated to websites of the 16 sources to find security and privacy advice; we used the search functionality when the advice was not easily

findable and used keywords like: “UPI”, “security”, “safety”, and “tips”. We found that the advice on these websites used a variety of names and categories (Table 3).

Participant Recruitment.

We recruited 11 participants from our personal networks and then used snowball sampling to recruit 15 participants from their social and professional networks [40]. We selected participants who used UPI and were willing to share their experiences with UPI in a 30-minute interview. We primarily used mobile and WhatsApp calls and messages to contact participants and sometimes used email if it was their preferred mode of communication. We recruited participants from seven states in India: Andhra Pradesh, Telangana, Karnataka, Maharashtra, Uttarakhand, Delhi, and Gujarat. After 18 interviews, we discussed participant diversity, and recruited eight participants with non-tech educational/employment backgrounds. Our sample, while of course not representative, was diverse across India with participants from seven states.

Pilot Testing.

We conducted two pilot interviews with colleagues who have used UPI to ensure the validity, comprehensibility, and language of the interview questions. We used the feedback from the pilot interviews to refine the interview guide for better clarity and flow of interview questions.

Interview Procedure.

We conducted 26 semi-structured interviews using Zoom and WhatsApp calls, depending on the convenience of the participants. The interviews lasted 20 minutes on average and were always led by the lead researcher, who spoke the languages participants were most comfortable with—Telugu, English, and Hindi. We sent participants a consent form and study description prior to the interviews; once they responded with interest, we scheduled interviews. Additionally, we received verbal consent to record and transcribe the interviews using Zoom at the start of the interview. We used Zoom to record and generate transcripts since it provided transcription support for all our interview languages, we used Zoom to record WhatsApp interviews as well. The lead researcher manually fixed the errors in the generated Zoom transcripts.

Interview Guide Structure.

The lead researcher created the initial interview guide and iterated the guide based on feedback from the rest of the team. The interview guide is structured in five sections: a) Participant’s UPI Background, b) UPI Usage Questions, c) Information Sources and Learning, d) Concerns and Challenges, and e) Safety Practices and Awareness; the complete interview guide is available in Appendix A.

Participant Demographics.

Our 26 recruited participants had diverse work and educational backgrounds (Table 1). Most participants had at least a bachelor’s degree; a few did not have a degree; some worked in non-technical fields. The average duration of UPI usage

Table 1: Participant Education and Work Experience.

Participant	Education	Work
P01	Computer Science Engineering	IT Engineer
P02	Computer Science Engineering	IT Engineer
P03	Bio-Technology Engineering	Student
P04	Computer Science Engineering	IT Engineer
P05	Computer Science Engineering	IT Engineer
P06	Computer Science Engineering	IT Engineer
P07	Computer Science Engineering	IT Engineer
P08	Electronics Engineering	Tele-Communications Engineer
P09	Metallurgy Degree	Student
P10	Medical Degree	Student
P11	Commerce Degree	Journalist
P12	High School	Home Maker
P13	Commerce Degree	Medical Agent
P14	Computer Science Engineering	IT Engineer
P15	Arts Degree	Factory Employee
P16	Pre-University	Business
P17	Electrical Diploma	Electrician
P18	Computer Science Engineering	IT Engineer
P19	Commerce Degree	Retired
P20	Commerce Degree	Accountant
P21	Commerce Degree	Retired
P22	Management Degree	Economics and Management
P23	Pre-University	Student
P24	Bio-Technology Engineering	Student
P25	Commerce Degree	Home Maker
P26	Electrical Engineering	Railway Technician

among participants is 4.5 years, indicating a prolonged and familiar use of these applications since the launch in 2016 [3].

PhonePe [6] was the most used UPI application among participants, but they also used Google Pay [7], Paytm [41], and BHIM [42]. This choice aligns with Statista’s findings that PhonePe holds the largest market share in UPI usage [23]. Users used UPI for transactions from the minimum of one Indian Rupee (INR, approximately \$0.012) to the daily upper limit of one lakh INR¹, highlighting its ubiquity across small purchases and larger transactions [43].

4.2 Data Analysis

Here, we explain the data analysis process we followed for the advice analysis and semi-structured interviews.

Advice Data Analysis.

We analyzed security and privacy advice from official UPI entities, banks, and regulatory bodies we collected using content analysis [44]. The primary coder coded all the advice and created an initial codebook consisting of nine main codes and 28 subcodes. The research team verified the codebook and coded data in multiple sessions. We then used affinity diagramming [45] to organize the codebook and identified six categories that we use to report our results (Section 5.1).

¹Lakh is a unit in the Indian numbering system equal to one hundred thousand 100,000 and the scientific notation is 10^5 ; approximately \$1,150

Rather than seeking data saturation for advice analysis, we focused on curating a list of security and privacy advice provided by popular banks and UPI providers.

Interview Data Analysis.

We used thematic analysis to interpret interview data [46]. Initially, the primary coder independently coded 18 transcripts using in-vivo coding and memoing and developed a codebook closely advised by two other authors [47]. We iterated and condensed the codebook through discussions. We then discussed the codebook with the research team and iterated the codebook to improve code definitions and consistency. To ensure the reliability of the codebook, a secondary coder independently coded 20% and reached an inter-coder agreement of $\kappa > 0.8$ [48]. The primary coder used the codebook to code the remaining eight transcripts. The researchers then met as a team to generate themes from the coded data. We identified themes in six categories that we use to report our results (Section 5.2). Our codebook is appended in Table 4.

Positionality statement.

In line with thematic analysis, our subjectivity was instrumental in collecting and analyzing data and presenting findings. Two researchers who are originally from India have experiences using UPI themselves. These experiences enabled us to reflect on various security challenges users encounter and were resourceful in creating the interview guide and data collection. Also, the interviews were conducted by a researcher who could speak the participants' native languages, enabling us to engage in deeper conversations with them.

4.3 Limitations

Our study has several limitations. First, while our study provides valuable insights into security and privacy perceptions of UPI users in India, due to the qualitative nature of our study, we may be subject to biases such as recall bias, observer bias, and participant self-censorship [49], [50]. Also, cultural and social associations with digital payments and privacy concerns may vary significantly across different populations in India, which could influence user perceptions and behaviors that were not captured by our study. Our recruitment was limited to convenience and snowball sampling, therefore, we may not have been able to reach a more diverse sample from all over India. However, we tried to recruit as diversely as possible within our reach and only stopped recruiting after we achieved data saturation [51], [52]. Our sample was skewed toward young and educated participants, however, it entailed participants without strong technological experience. We do not claim that our results are representative of the entire Indian population.

Another limitation is that we selected banks and UPI providers for advice analysis based on prior research and surveys [1], [53]. We also relied on prior work to identify the most used banks and UPI applications in India, potentially excluding a few other banks and entities that also offer

UPI safety advice [1]. As a result, our analysis may not fully capture the breadth of guidance available to UPI users from various banks and entities. Further, our interview coding process maybe influenced by a subjectivity bias, as the majority of the transcripts were coded by a single coder. To reduce the impact of subjectivity in interview data analysis, we regularly discussed coding within the team and double coded 20% of the transcripts, and we recoded transcripts based on disagreements and changes made to the codebook.

5 Results

We report on results from our analysis of official UPI security and privacy advice (Section 5.1), and user security concerns and challenges, together with security behaviors and rationales regarding UPI security (Section 5.2). We compare and contrast user reports with advice in official sources, specifically linking whether and how official advice covers concerns/threats that users report. Our results indicate that participants are actively implementing security measures, such as not sharing PIN, verifying the receiver details before a transaction, not clicking on unknown links, and being aware of scams and frauds associated with UPI. They reported that they generally receive UPI-related security and privacy advice through word of mouth, social media, and advertisements on television. Even though official UPI bodies, banks, and the government provide security and privacy advice, our results reveal that not all safety recommendations published by official entities on their websites effectively reach UPI users.

5.1 Official UPI Advice Analysis

We describe results from analyzing 16 online advice documents, scraped in 2023 from public and private sources (Table 2), including India's top ten banks, four UPI applications collected from Statista's survey [20] and previous research [1], and two regulatory bodies NPCI and RBI. We categorized advice in six major categories that emerge from our analysis, *a) Transaction Security*, *b) User Data Protection*, *c) PIN/Password Protection*, *d) Device and App Security*, *e) Customer Support and Reporting Sources*, and *f) Handling Unknown Situations*. We discuss the sources, content, and prevalence of 16 sources (in parentheses) of the advice.

Transaction Security Advice.

Transaction protection advice is provided by entities specifically related to UPI transactions, along with prevalence (in parentheses) from our sample. The advice includes practices like *entering the UPI PIN (13)* and *scanning QR codes (9) only for sending money* and *only on official UPI pages (3)* to avoid scams where fraudsters trick users into entering the PIN on suspicious pages to receive money instead. Before approving any transaction, users are suggested to *understand the payment request details (5)* and *verify the type of transaction being performed (5)*. It is also advised to *verify the recipient's*

UPI ID (9) and Virtual Payment Address (2), checking the receiver's details (7) and the amount being transferred (2) before proceeding with transactions; advice warns that *transactions once made cannot be reversed* (3). During transactions, users are recommended *not to close the window abruptly* (5) to confirm completion. *Regular monitoring of bank statements and transaction history* (5), along with checking SMS and alerts (4) after transactions is recommended to verify the debits and credits in the account, and be alerted to unintended transactions. To safeguard sensitive information, users are suggested to *make transactions privately* (2) and after completing a transaction, *log out from the UPI app* (5) to prevent unauthorized access. If a transaction request sent to users seems suspicious, it is recommended to *reject it immediately* (4) and in case of potential fraud, users are advised to *block the UPI and their bank accounts* (4).

User Data Protection Advice.

Entities provided many key suggestions for UPI users to protect their data which include *never share personal, banking (11), contact or social media details (3) via calls, texts, and emails (5) even with banking officials* to protect their privacy and prevent potential fraud. UPI users are also suggested to always *check official emails from banks (3) and verify the authenticity of the sender (2)* before responding to any requests for information to avoid falling victim to phishing. To protect their sensitive information users are recommended to *avoid saving banking credentials on their phones (8) and sharing the device with strangers (2)*. When in public places, users are recommended to *avoid entering their PIN or password (3) or saying it out loud (2)* to prevent it from being observed or overheard by others. The advice also includes *avoid clicking on links for KYC, PAN, or Aadhaar updates (1), offers related to the caller or dialer tunes (1) and checking USSD (Unstructured Supplementary Service Data) requests thoroughly (1)* as these are often used in phishing attacks. *Managing their own banking accounts (3) and checking the time of login every time they access (1)* their accounts is recommended to ensure account security. *Being aware of phishing (1)* and how to recognize them is also suggested for users to avoid falling victim to such attacks.

PIN/Password Protection Advice.

UPI users are advised by entities to protect their *PIN, password, OTP, generated codes, and CVV by never sharing with anyone (14)*. They also suggested *creating a strong unique PIN/password (6) by including random characters (1)* to prevent attackers from cracking it easily. While creating a PIN/password it is recommended to *avoid the inclusion of personal details (3) and have a reminder to change it frequently (6)* to ensure safety. After creating a PIN/password users are advised to *memorize it (1) and set the maximum number of incorrect PIN/password attempts to three (1)*. Instead of having the same password for multiple accounts, users are suggested to *have different credentials (3) and disable the auto-fill option (1)* to avoid unauthorized access by sharing

the device.

Device and UPI Application Security Advice.

Device and app security advice includes safety steps like *avoiding public Wi-Fi networks (6), Bluetooth connections (2) and devices (3)*, which can be insecure and restrict using *third-party screen share apps (6)* which can exploit the device safety. Users are suggested to *enable only necessary permissions (3) and disable the GPS feature (1)* to limit exposure. To protect against attacks and be updated with the latest security threats, entities advise users to *keep their devices (2) and antivirus software updated (4)*. It is also recommended to *be aware of the dangers of SIM cloning or swapping (3)* and in case of a lost or stolen device, *log out from apps from the web applications (2) and block the SIM immediately (2)*. To protect the device and app from unauthorized access users are advised *not to leave the device unattended (2), and not allow their browser to save card or account passwords (1)*. *Setting up account notifications (2)* is advised to monitor for suspicious activities and users are recommended to *notify banks if they change mobile numbers (2)* to ensure continued access to their accounts. To avoid security threats users are advised to *block the suspicious contacts, change default admin credentials (1) and not to use jailbroken or rooted devices (1)*. It is recommended to have *strong authentication methods (1) and keep track of unused apps and remove them (1)*. To prevent skimming attacks while making transactions, users are advised to always *check for unknown devices in ATMs locations (1)*.

Customer Support and Reporting Sources.

Customer support and reporting sources advice includes safety measures such as *using secured websites with HTTPS encryption (5), and trusted sources like Google Play/Apple store (8) to download, and install the applications*. Users are also suggested to *use valid helplines from official websites (11) instead of public helplines (6) to report unauthorized transactions (7)*. *Reading and saving reviews (1), and terms and conditions (1) of the apps* are recommended to understand the usability, security, and privacy-related information. In case of lost or stolen devices, users are advised to *contact the bank and law enforcement authorities (4)* immediately to minimize the risks.

Handling Unknown Situations.

Entities recommend users *avoid clicking on unknown links (10), email attachments (5), and responding to unknown emails, texts, calls (7), and websites (5)* to mitigate scam attempts. Users are also advised to *be cautious when coming across suspicious apps or cashback offers (9), as well as unknown merchants (5)*, to prevent potentially fraudulent transactions. To protect user privacy they are suggested to *prevent seeking advice from unknown sources (4) and stay away from distractions and demands (3) and unsafe places in unfamiliar situations (3)*. It is also recommended to *avoid using unknown public charging ports (1) and not forward*

unsolicited information (1) and carefully check for typos, and grammar errors in emails (1) to prevent phishing attempts.

5.2 UPI users' perceptions, behaviors, and awareness

We discuss results from the interview study, exploring UPI users' perceptions, concerns, challenges, behaviors and advice sources participants mentioned receiving related to UPI security.

5.2.1 Participants' security concerns with UPI and their mitigations

We report participants' concerns and perceptions around UPI security. In the interviews, these were closely interwoven with mitigation behaviors: participants often reported concerns, and how they behave to minimize the concern. We therefore report these closely interwoven results in conjunction, and refer back to similarities with official advice, where appropriate.

Participants express distrust towards UPI apps and prefer to use UPI services from their banks.

Concern: Participants expressed a distrust towards UPI apps, because they are a 'Middleman' (P26, Railway Technician) acting as an interface between the user and the bank; therefore, they are unsure how stand-alone apps handle issues they may encounter, specifically around faulty transactions and scams.

Mitigation: To avoid issues with stand-alone UPI apps, some users mentioned switching to in-bank UPI services (like iMobile [54], YONO SBI [55], and AXIS Pay [56]). Participant P22 explains their trust towards banks by emphasizing that their banks have a better regard for their security and privacy "When your bank is quite active regarding your privacy and your security, you know your account is in the right hands, and you also are kind of motivated" (P22, Economics and Management). Further P26, who also switched from a stand-alone UPI app to a in-banking UPI service—even though UPI apps have a "better" user interface—mentioned that they trust banks to deal with their concerns and issues over stand-alone apps, "User Interface wise PhonePe (stand-alone app) is better than the bank app, but security wise, I always prefer the bank-integrated UPI app. If some problem comes up, we can directly put up the issue with the bank, so they will resolve it" (P26, Railway Technician). Participants prefer banks and their apps over stand-alone UPI apps even with similar or better functionality because they are accustomed to and trust their banks. This mitigation is not directly mentioned by official sources (Section 5.1), but more vague advice in official resources that advises to only work with trusted sources is present.

Users conflate scams with UPI platforms.

Concern: Some participants reported that they—mistakenly—associate new digital scams with UPI. In (rural) areas without

coverage by actual physical banks, Customer Service Points (CSPs) [57] act like mini banks, and assist customers with money withdrawals and deposits using biometric fingerprints and Aadhar numbers². These CSPs are associated with government initiatives but are established by private individuals, often store owners, where bank account holders can visit and utilize regular banking functionalities [58]. Participants reported instances of fraudulent activities, where users' funds are debited from their accounts without their consent, particularly after utilizing biometric details at these CSPs and government-subsidized shops. As P12 recollects, "When people went to subsidized shops and used thumb authentication, their money got deducted. I heard many people losing money this way. Few lost all money from accounts" (P12, Home Maker). Participant P13 also says "Recently I heard many incidents related to CSPs where money gets deducted from the account without user initiation. They collect biometrics from customers to provide easy access to money withdrawals etc. All these are related to PhonePe and Google Pay majorly" (P13, Medical Agent). Both these participants believed that fraudulent actors at the CSPs misuse biometric data they collect and generally specified being suspicious towards UPI apps, even though these cases of fraud are unrelated to UPI.

Mitigation: To circumvent issues with CSPs, and scammers, participants reported avoiding using biometric authentication, and decreasing use of UPI: "So I am using less of these apps and not providing my thumb authentication anywhere" (P13, Medical Agent). This issue is not addressed in official advice (Section 5.1).

Concern: One of the major concerns participants brought up was with adversaries tampering with payment links and QR codes, which can result in users losing money. Since UPI payments to merchants can be made by scanning a QR code or clicking on a link participants worry about the authenticity and safety of the links: "whenever I try to buy something via online and opt for UPI payment, they will send some link where I need to input my OTP" (P04, IT Engineer). Participants state that the source of many of the scams around UPI is entering sensitive details on the unknown/uncertain links sent to users. Additionally, they also mentioned worries about the validity of QR scanners in markets or outside settings, "...overnight, without the shopkeeper's knowledge, those QR codes can be changed or stuck upon other QR codes. So, whenever we make the payments, it could be going to the right person or to the wrong person, which we won't be aware" (P01, IT Engineer). Further, participants mentioned scammers making users reveal their PINs and passwords while pretending to be bank officials. As P13 explains, "I also heard from many that they get calls telling their Adhar is not linked to the bank account or KYC (Know your customer) is not done, so repeat the OTP to link the account. Once we tell the OTP, the entire amount in the accounts is being deducted" (P13, Medical

²Aadhar is a unique identification number issued by the Unique Identification Authority of India (UIDAI) to residents of India

Agent). Prevalence of scammers instills fear in UPI users resulting in some users rejecting UPI apps altogether—as in some cases even reporting scams is unhelpful.

Mitigation: Participants P07 and P09 mentioned trying to double check and verify whether they are paying the right person, and also verifying whether payment links are legitimate: “I’ll cross-check with them twice with their name and number, like while crediting money to someone” (P07, IT Engineer). Participant P04 mentioned trying “to clarify or check particular links, and details present on that web page, and cross-check whether that link belongs to either Amazon or whatever website I’m paying for”. Participant P08 also said they verify when they enter their PIN “we enter this PIN only to send money, but we do not enter it to receive money” (P08, Tele-Communications Engineer). Carefully checking links and QR codes is part of official advice (Section 5.1).

Cybercrime complaints are mostly unreliable and unhelpful.

Concern: When users encounter scams or fraudulent activities in UPI transactions, they typically file complaints under cybercrimes. However, participants mentioned these complaints often remain unresolved for extended periods, and the recovered amount is usually only a partial sum of the total lost funds. Participant P11 recounted their friend’s experiences where, “they got a partial amount but not completely, and yes the complaints are still open in police stations” (P11, Journalist). Participant P12 also heard of many complaints that local residents filed in police stations but could not recover much: “the issues didn’t get resolved even when they made a complaint in the police station. They could recover only limited money” (P12, Home Maker). Participant P13 expresses their concern about how banks can track the transaction flow easily yet they are not able to track the scammed money and recover it completely, and says “there seems to be an issue with the banking sector and they are not helping users who lost money. When we complain the support is not enough, they file as cyber crimes but most of them are not helping users. My cousin also lost 30k INR” (P13, Medical Agent).

Mitigation: We did not encounter any specific mitigation strategies participants followed for filing cybercrime complaints or retrieving lost amounts, however, they hinted towards not using UPI or keeping their phone on them at all times to avoid misuse (P10, P12). Though this concern is unrelated to the UPI platform itself, such encounters with scams and unhelpful follow up leave users feeling helpless, and contribute to a general distrust in UPI apps, and worse rejecting the use of UPI apps. This problem is not adequately addressed in official advice (Section 5.1); customers are only recommended to report crimes, but not advised how to follow up or solve the problem.

Participants worry about security risks of leaving devices at repair shops.

Concern: Participants expressed concerns about handing over their mobile devices to repair shops, particularly when dif-

ferent applications are installed on the device and sensitive information is stored on them. Participant P01 expressed they fear the potential risk of unauthorized access: “Ideally, the person in the repair shop knows more than what we actually know. If he has good technology in his hands, he could even hack into our phones, and log into our phones. So that’s a concern when I give the phone to repair” (P01, IT Engineer).

Mitigation: To overcome this, participant P12 only uses services of repair shop personnel they are familiar with: “I only trust to give the phone to known people/shops. We do not give to shops which we do not know” (P12, Home Maker). Participant P16 has their son watch over the device while it is being repaired: “My son stayed there till they fixed the issue and got the mobile back” (P16, Businessman). Additionally, for UPI transactions, the Subscriber Identity Module (SIM) plays a crucial role in verifying the mobile number associated with the transactions; therefore, removing the SIM before handing the phone over to repair shops or blocking the SIM if the phone is lost can help protect against unauthorized UPI transactions. Few participants state this, for example P11 describes: “I always take off SIM from the mobile before giving for repair. So they won’t get our data or any information after SIM removal. Also without SIM, they can’t make any UPI transactions” (P11, Journalist). Other participants learned this through participating in our study: “Thank you so much for this advice, I didn’t know that removing the SIM is a security measure” (P25, Home maker) and also a few did mention they would take these precautions.

Further, P08 and P04 who were iOS users reported a feeling of security that they associate with their iPhones. Participant P08 mentioned that the mandatory use of face ID to access apps helps them think it is more secure: “first time, like when once you minimize or close your app and open your UPI application, you need to use face ID to have this application open. I use face lock mostly because there is no finger authentication on iPhones. It is either a face lock or a PIN/Password. Also, I’ll just lock my phone and give it to repair. It’s not that easy to crack iPhone” (P08, Tele-Communications Engineer).

Participant P04 mentioned that using 2FA to access UPI apps on their iOS device makes them certain that the repair shop personnel cannot misuse the device: “I do not think they can break the initial password of my iPhone, even if they break also, I do not know, whether they can get into my PhonePe or Google Pay apps. It has 2-factor authentication, with my face ID, I do not think they can enter into those apps” (P04, IT Engineer). Our findings are consistent with those reported by Kostan et al., who also qualitatively observed higher trust in iOS devices’ security and privacy [59]. Blocking SIMs after device loss is mentioned in official advice sources (Section 5.1), but repair shops are not addressed specifically beyond advice to “be cautious” when handing over devices. Official advice includes locking UPI applications.

Participants worry about losing their device but are familiar with follow-up.

Concern: Some participants pointed out concerns around losing their device, with UPI and banking applications installed in it, but most participants who shared the concern shared several steps they take to protect their banking information.

Mitigation: Having first-hand experience with losing a mobile with UPI applications in it, P01 said “...the immediate measures which I took was to contact the customer care or the banking department or the bank to please block all the transactions, which should happen until I give a consent or I buy a new phone and install the app and start using the UPI. So until that time, I had requested to block all the transactions incoming and outgoing from my account” (P01, IT Engineer). Banks always provide different options to contact them to report issues like visiting banks in person, and contacting customer support, and every entity always has an emergency contact to report banking frauds quickly and efficiently. Even when not having previous experience with losing their mobile or seeking assistance on how to handle such a situation, P10 and P12 mentioned that “I think I’ll just call my bank and block the card or block the transactions” (P10, Medical Student). “We should go to the bank and inform them to block the transactions. That’s all we know” (P12, Home Maker). The majority of the participant’s first action is usually to contact their bank immediately and request to block or freeze their account to prevent any unauthorized transactions. Participant P15 also mentioned filing a complaint at the local police station, providing details about their lost mobile to track the device, saying “Contact the police station and also disable all the apps by web login. Also, contact banks to block the account” (P15, Factory Employee). Only very few participants even mentioned going a step further by blocking the SIM card or deactivating the mobile number associated with the lost mobile. Official advice recommends blocking SIMs after device loss (Section 5.1).

Auto debit feature in UPI exacerbates safety concerns.

Concern: Many participants mentioned that they avoid using the auto-debit feature in UPI apps which suggests that users are hesitant about letting money be automatically deducted from their bank accounts. They also express fears and reservations regarding the auto-debit functionality.

Mitigation: Participants mentioned circumventing this fear by not opting into auto-debit features “I do not opt for this auto deduction thing from the UPI app. I do not prefer or like it, I’m not sure how those things work. I do not feel it’s secured. So I do not opt for that option” (P04, IT Engineer). Participant P06 still uses this feature but limits it to only a few and not for important types of transactions, “I do not use it for all the things it’s just for entertainment subscription purposes like Amazon or Netflix, etc” (P06, IT Engineer).

Interestingly, P08 is unaware of an auto-debit feature in UPI, saying “No, that’s actually not possible. You have to enter your 4-digit UPI PIN to make a payment and payment reminders are actually not related to UPI. But I think it’s the applications that keep reminding you when you pay

any electricity bill using your meter number” (P08, Tele-Communications Engineer). This indicates that UPI providers should address misconceptions and educate users about the benefits and safety measures associated with using auto-debit options. This is not addressed in official advice (Section 5.1).

General security behaviors.

In addition to mitigating specific security concerns, participants also reported general security behaviors, without explicitly naming threats of e.g., scams or shoulder surfing. All participants mentioned a few to many safety measures that they follow. They highlighted not sharing PINs with anyone: “I will remember PIN carefully and I will not share with anyone other than my son” (P16, Businessman). Many also mentioned changing their PINs regularly, not writing PINs down, and avoiding using familiar dates as PINs. In addition to PIN-related precautions, P12 also underlined the need to be cautious of sharing any sensitive details: “Even if they claim that they are calling from banks, I do not share any sensitive information” (P12, Home Maker). Several participants also discussed how they stay safe in public places while using UPI applications like P11 says “I haven’t used free wi-fi and also I avoid transferring money in those scenarios” (P11, Journalist). Most of the participants like P12 also followed other strategies like “...not giving the phone to anyone and not doing transactions in public. We do it mostly at home and do not use the PIN-related numbers in public” (P12, Home Maker). Surprisingly, P10 specified that other than not sharing PIN with anyone they do not follow any other safety measure says “PIN is obviously the first thing, we do not disclose it. I think other than that, I do not follow anything” (P10, Medical Student). Many of these behaviors are mentioned in official security advice (Section 5.1).

Participants follow default security settings on UPI apps, unaware of other safety settings.

Participants mostly report using default settings, even though they can be updated with additional security measures like enabling two-factor, biometric authentication, and enabling notification alerts for any suspicious activities. Participant P09 said that they left the settings to default from the beginning, saying “I think mine’s on default, I didn’t change anything” (P09, Metallurgy Student). Most participants generally believe that the default settings are secure enough, while others who want to be “safe” also mentioned updating their default settings with 2-factor authentication and other authentication techniques: “I have 2-step authentication for my Google Pay app. For opening the app, you’ll have security authentication and for the payment, you’ll have one more step of security authentication” (P10, Medical Student). Further P10 and P12 also mentioned keeping their phone on them at all times for more safety and the fear of misuse. Other participants also mentioned changing their PIN regularly and enabling fingerprint and face biometric authentications for UPI payments, and they expressed a general lack of awareness for other safety features. For instance, P12 relies on their kids to set up UPI

payments on their app and are unaware of any other safety features. Features they did not bring up include setting up different UPI IDs for separate bank accounts, in Google Pay [60] and PhonePe [61] apps, such as not using a default UPI ID and instead configuring distinct IDs for each bank account. Likewise, no participant mentioned the settings from Google Pay [62] and PhonePe [63] apps that allow users to customize their unique UPI number to hide their original mobile number for safety reasons.

Biometric authentication for UPI feels more secure than other authentication methods.

Biometrics (both finger and face authentication) holds importance as it is unique to each user and cannot be easily stolen compared to PIN, Passwords, and Pattern locks which can be overlooked or remembered [64]. The majority of the participants agreed with its significance, where P21 said, *“compared to PIN, biometric is safe because it can’t be stolen and it is always with me”* (P21, Retired). In case of stolen or lost mobile P26 mentioned that *“Because sometimes we feel like somebody can look over your password, they will identify a pattern also, and when your phone is stolen or lost and goes to some person he/she may open and use it. So that’s the reason why I use biometrics for it”* (P26, Railway Technician).

Participants prefer PIN-based transactions over “UPI Lite” feature for security and control reasons.

Participants expressed a lack of interest in the new “UPI Lite” PIN-less transaction feature highlighting preference for PIN-based transaction control. Participant P11 said *“No, I won’t feel safe with transactions without entering a PIN and not being able to check the details of the receiver before the transaction. Without a PIN anyone who gets access to our phone can do transactions, so I do not feel safe at all”* (P11, Journalist). This indicates the preference for familiar and payment methods with perceivable security processes, possibly due to concerns about the safety of transactions.

Participants prefer PIN authentication over pattern locks for safety reasons.

UPI offers various options for users to secure the application, including pattern locks, biometric authentication, and PIN locks, allowing users to choose the authentication method they prefer. Participants mentioned that they feel cautious about using a pattern lock for authentication: *“...patterns are very easy to decode like you put the pattern, and they can identify what the pattern was very quickly. When it’s PIN it doesn’t show the number”* (P03, Bio-Technology Student). They also mentioned that PIN is preferred because PIN digits are masked while entering and shuffled and changed each time the user attempts to log in, reducing the likelihood of guessing the PIN typing pattern.

5.2.2 Participants’ sources of UPI security information and awareness

We explore information sources reported by participants to learn about UPI functionalities, safety measures, and handling issues.

Participants recall receiving limited and basic advice from banks and UPI providers.

The majority of participants recalled only a few pieces of advice from official sources, such as not sharing one-time passwords (OTPs) or PINs sent via mobile text messages or notifications. Participant P05 recalled *“never share your OTP or never share your PIN with anyone else. These are the most reminders or notifications that I have seen”* (P05, IT Engineer) and another stated *“One constant message I always receive is, do not share your OTPs. That’s the main thing”* (P10, Medical Student). However, our analysis in Section 5.1 revealed that banks and UPI providers offer a wide range of other advice through advertisements, campaigns, official websites, text messages, emails, and calls, particularly for high-value transactions. However, most of these additional advice or the means through which they are conveyed were not mentioned by participants. This suggests that the advice may not be reaching everyone as intended by banks and UPI providers.

The sources users follow for advice differ from their expectations from banks and UPI providers.

For UPI advice most of the participants mentioned searching online, browsing, and seeking advice from family and friends like P06 *“I think I got this from my parents because they usually say that since you do a lot of transactions, make sure that you change your passwords frequently enough, and it is safe”* (P06, IT Engineer). Participant P11 specifically mentions the sources they follow, says *“I search on Google and YouTube channels”* (P11, Journalist). In a similar way participant P08 also discusses the forums available online, says *“we can use Google like you have these forums or discussions where you can see the information too”* (P08, Tele-Communications Engineer). Whereas the platforms they expect banks and UPI providers to utilize for promoting safety and security advice are mostly TV advertisements, text messages, and social media. Few expressed doubts, suggesting that texts would be ignored *“Most of them do not read messages usually, if we get automated calls with safety measures they will at least listen for some time. TV advertisements are also better but YouTube ads can be skipped so not much use with it”* (P17, Electrician). Here they also mentioned that calls from entities could have a stronger impact since users cannot easily skip or ignore them, unlike YouTube ads or text messages.

Users feel safe and comfortable following safety steps discussed with friends and family.

Most participants mentioned that they contact their family and friends for suggestions, doubts, or clarifications. Participant

P12 mentioned *“My kids told me, as I am not very educated they helped me know all this information”* (P12, Home Maker). Participant P26 got their scams-related safety measures from *“family and friends, they share with me if they get to know any information”* (P26, Railway Technician). Discussing and sharing UPI information among each other helped: *“My source of information or safety measures are my kids, and also when we go out together with friends, we talk about somebody’s experiences like in the family or others. When we discuss more, it becomes easier to understand it better”* (P19, Retired).

To resolve issues, some users directly visit banks.

Participants without much experience with digital technology often choose to visit their bank in person to address any issues they come across: *“I will not contact anyone other than the bank, that’s my trusted source to solve the issue. I had an issue related to PIN once, the bank solved it. I never contacted customer support”* (P16, Businessman). This behavior suggests a preference for traditional, in-person interactions when resolving issues, rather than relying on online support. Most also do not prefer contacting customer support, especially participants with non-technical backgrounds: *“We do not take help from anyone other than my kids and bank. Only these two sources and we won’t go for the third option at all”* (P12, Home Maker). This highlights that users are relying on trusted and familiar sources to solve the issues with UPI, including close family, which safeguards them from falling prey to fraudulent advice and information.

Some users prioritized verifying security advice sources on social media, while others are confident.

Among the participants, there is a difference in the approach towards validating the security advice sources on social media platforms. Some participants prioritized verifying the trustworthiness of security advice sources before accepting or implementing the advice, while others did not worry much about checking the source’s credibility, as exemplified by P06: *“...whenever I come across this kind of new UPI update, I would check it in the application first, whether this is true or not, and if it is available to us, then search articles on Google. So I just cross-verify”* (P06, IT Engineer). Similarly, P22 says *“I check credibility through the followers someone has on YouTube. If they have 10 million views or followers and if I have been following that person, then, of course, I’ll take the opinion”* (P22, Economics and Management). Contrarily, P11 says they are less worried about the source of advice, *“I will look into any channels which showcase the information, I will also look into channels of motivators/influencers as well, wherever I get useful info”* (P11, Journalist). Participant P15 mentioned they make sure to trust only official channels for advice *“I trust only news channels and official channels for advice. Not any random information from anyone unknown. I trust major news channels”* (P15, Factory Employee). Many official entities (Section 5.1) highlighted that it is crucial for UPI users to verify the authenticity of advice on social media to avoid any misinformation or scams that can lead to

financial loss or security breaches.

6 Discussion

We explored UPI users’ perceptions of privacy and security, their mental models, challenges, concerns, and other key aspects through 26 semi-structured interviews. Additionally, we conducted an analysis of advice provided on the websites of 16 official UPI sources, categorizing the advice. Our findings indicate that, although authorities offer extensive safety advice, users follow only a limited subset of it. To better understand this gap, we assess how users mitigate their concerns, and compare and contrast with official guidance, finding mismatches. In this discussion, we examine the implications of our findings and recommend improved methods for communicating safety advice and best practices to UPI users.

We find that participants report concerns and mitigate those in ways that feel, more or less, secure to them (Section 5.2.1). We however observe that not all threats mentioned in official sources (Section 5.1) are also brought up by participants, and, consequently, participants also do not protect from these threats. Participants mostly create PINs (as promoted through the UPI or banking apps), and use biometric authentication when prompted. They also use physical measures (such as keeping their phone on them). However, general vague fear of scams and UPI leads to users choosing to use banking apps over dedicated UPI apps, accepting a less task-appropriate user interface in exchange for the trust they place in banks over UPI. They also learn to check links and OTPs multiple times; however, the effectiveness of unspecific worry is unclear. We further observe that the somewhat exhaustive advice is not accessed by participants (Section 5.2.2), who learn through video formats or unofficial advice sources, as well as friends, and do not report accessing official advice websites, an observation that tracks with prior research [65].

We think that there is an opportunity to close the awareness gap for users, banks, and UPI providers, and give recommendations to these actors below.

6.1 Recommendations

We offer recommendations to address the security and privacy threats that come with UPI. When these recommendations come directly from our data, we indicate this by backlinking to the respective subsection. We also offer recommendations that more abstractly follow from our data, and the experience of collecting and analyzing the data, combined with our expertise as security researchers.

Recommendations for UPI users.

We are aware that “demanding more from users” is generally not a promising strategy to increase their security and make recommendations mostly for other stakeholders [66]. However, in the absence of a better ecosystem, we do make recommendations on how UPI users can be proactive about

UPI security: To safeguard themselves from misinformation, users are encouraged to read and follow the safety and security suggestions provided by UPI providers and banks on their official channels, websites, and other mediums (Section 5.1). Additionally, exploring all the available features and settings in UPI apps can enhance the overall UPI experience. Instead of just relying on familiar security settings, users are encouraged to explore new settings and try them out under supervision for safer transactions [60]–[63]. Users have to be mindful about linking their biometrics data and sensitive details outside of banking and other government-related entities for their safety. Another suggestion for users is to be cautious with contacts pretending to be government officials and follow their guidance for any procedures. It is important to always verify the source from which we seek guidance by double-checking if they are from official sources, verifying through multiple sources, and when in doubt contacting the official customer support for confirmation.

Recommendations for banks.

Banks should help users know about all the security options in UPI apps, not only biometrics (often bypassable) and passwords. They should offer simple and clear instructions on configuring settings that can enhance their safety. Also, they have to educate UPI users, especially those in rural areas, about the differences between UPI transactions and scams linked to other banking services like CSPs. They should highlight to users that none of the entities contact them for any sensitive details and they should avoid sharing the details under any circumstances. As banks have a physical presence in all parts of the country, they should increase awareness by running campaigns and advertisements, as well as counseling customers during their visits and they should promote more information regarding safety and privacy (features) through other mediums like calls, messages, and emails to reach all kinds of populations—however, we are aware that these may be the exact channels used for misinformation. Banks should also advise users on safety measures when leaving devices at repair shops, handling server downtime, and protecting SIMs, which are critical to UPI transactions. Documenting common UPI scams and promoting awareness, especially among rural populations and young adults, can help reduce vulnerability to them. Regular campaigns in schools and institutions can educate young adults about financial safety, and continuous updates during bank visits on the latest security features and scam alerts can further protect users.

Recommendations for UPI providers.

We suggest that UPI providers make safety advice more accessible and understandable for users by utilizing various communication methods, such as videos, tutorials, text messages, and pop-up alerts within the app. Also, it would be helpful to clearly highlight all available safety features beyond sending and receiving money, such as having unique UPI IDs for different bank accounts, customizing the UPI number instead of using a mobile number, and demonstrating how these features

can benefit users in safeguarding their privacy and security. Providers should help users recognize potential risks when interacting with third-party service points and keep users updated with tips on improving mobile device security apart from basic locking mechanisms for better protection. As there is a diverse demographic using UPI, young adults are particularly vulnerable as they are in the early stages of financial independence and accessing banking and technology and so it is crucial to inform them about safeguarding their privacy and security. We suggest organizing information campaigns around heavily used platforms. We also recommend the addition of a usable “Report Transaction” feature for quick action on suspicious activities. An option for immediate transaction retrieval could also help users rectify unwanted transactions.

Recommendation: YouTube and social media for UPI security advice.

Perfunctory review of YouTube channels, social media, and TV advertisements indicates that both official sources, such as banks and UPI providers, and private entities maintain channels to promote fundamental security advice, like avoiding sharing OTP and PIN, not sharing sensitive information, not sharing device and being cautious with public transactions, also the most commonly recalled advice in our interview study (Section 5.2). However, the coverage of emerging threats, scams, and updates on new security features appears limited. To enhance the outreach of advice, these platforms should be used to disseminate advice on emerging scams and advanced security settings. Further research can examine in detail how and what advice is shared via YouTube, social media, and TV advertisements to gain insights into the effectiveness and reach of UPI security recommendations and their influence on user behavior and awareness.

Recommendation: Public security communication.

We think that there is an opportunity for effective communication with multiple goals: (1) teach users about threats; (2) teach users about mitigations; and (3) help users distinguish between threats to UPI and general threats of scams unrelated to UPI. We think that effective, in-time communication about widespread scams, their relation (or not) to UPI and mitigations can both keep users safer from scams, and also decrease sometimes unfounded fear in using ubiquitous UPI technology.

6.2 Future Work

Giving actionable advice that is applicable to end users is hard, and rarely done well—this is well-documented in security and privacy literature [65], [67], [68]; this also applied to UPI in a vastly different context than most security advice studies [69]–[74]. We recommend researching how to give actionable advice in this specific context, with the potential to improve financial security for hundreds of millions of users. In future studies, we recommend exploring policies implemented by UPI apps, including grievance procedures,

privacy policies, terms and conditions, and open-source licenses. By analyzing these policies, we may be able to better understand how these apps can be developed to improve the usability, privacy, and security of UPI users. Prior work has researched how businesses handle UPI payments [35], similar to our study of general users; application development seems a promising future avenue.

Further, it may also be helpful to analyze the systems and policy design ideologies behind UPI platforms. Research with developers and monitoring authorities could provide insights and information about how system design and regulatory frameworks impact the security and privacy of UPI transactions. Additionally, a follow-up quantitative survey with a representative sample of UPI users would help quantify security issues, including personal experiences with UPI scams and frauds. As UPI evolves, it may also be important to investigate the adoption and impact on security of new features like “UPI Lite”, particularly regarding PIN-less payments.

Finally, as UPI is gradually extending to other countries, conducting similar studies in international contexts could provide valuable insights into the perceptions and experiences of users outside of India. For example, the European Union has recently introduced a comparable instant payment system through the European Payments Initiative (EPI) [16], [17], [75]. Investigating such systems in different economies, mental models of how online transfers work, as well as different technological and legal landscapes can enhance our understanding of global payment systems.

7 Conclusion

Our study of UPI users and UPI security advice finds that while users demonstrate a certain level of awareness and attentiveness when conducting UPI transactions, there are noticeable gaps in their understanding of specific security features and misconceptions about associated risks. While users are cautious about certain aspects of transaction security, such as password protection and biometric authentication, their knowledge regarding other safety measures offered by UPI applications remains limited. This underscores the importance of targeted educational initiatives aimed at enhancing users’ awareness of the full spectrum of security features available to them, thereby strengthening their ability to mitigate potential threats effectively.

Through our examination of the official advice provided by UPI entities, we learned that there is a significant amount of information available from regulatory bodies, banks, and UPI applications’ official websites regarding safety and security measures. However, many participants are not utilizing these sources for guidance. Instead, they rely on alternative sources like online browsing, family and friends which seem to better meet their information preferences. Encouraging users to utilize official websites or exploring alternative sources that are commonly accessed could enhance overall security. This

finding underscores the importance for stakeholders and banks to promote the use of official websites as primary sources of safety advice or to consider alternative channels that are frequently utilized by users to promote safety advice efficiently. Overall, our analysis highlights the importance of improving the accessibility of official advice provided by UPI entities to ensure that users can easily understand and implement recommended safety measures, thereby enhancing their overall security when using UPI.

8 Research Ethics and Open Science

This section outlines our adherence to research ethics and our commitment to open science practices, contributing to the integrity and reproducibility of our research study.

8.1 Ethical Considerations

While our institution did not require ethical review, we closely adhered to principles in the Menlo report [76]. We de-identified all interview data and removed personally identifying information. Participants were compensated a study allowance of INR 500 (6\$) for their participation in the interviews. We informed participants that participation was voluntary and that they could withdraw from the study at any point in time. Since participants discussed their experiences with scams, we assured them that they could refrain from answering questions if they felt uncomfortable, and reassured them that we would de-identify their data and only report de-identified quotes. Additionally, since some participants were not familiar with Zoom, we offered to use WhatsApp calls or other technology they were most comfortable with. We also conducted the interviews in local languages, and participants felt comfortable articulating their thoughts.

8.2 Open Science

For the purpose of applicability and reproducibility, we attach our interview guide in Appendix A. To avoid re-identification of participants, we cannot share full transcripts. The details of entities and the advice extracted from their websites are available in the Appendix B (Tables 2, 3) along with the interview codebook used in our analysis in the Appendix C. Links to all the official websites with UPI advice we examined are included in the citations and can also be found in Table 3. We will also provide additional snapshots of these websites at the time of conducting this study upon request. Along with what we have included in the Appendix, we have also made the interview guide in English, Hindi, and Telugu, interview recruitment messages, consent form, and demographic questionnaire available at <https://doi.org/10.5281/zenodo.14773246>.

9 Acknowledgments

We would like to express our deepest appreciation to Anna Lena Rotthaler, Anastassija Kostan, Rachel Gonzalez Rodriguez, and Darya Zarkalam for helpful discussions and feedback throughout this research. We thank our participants for providing valuable input. We also thank all the anonymous reviewers and our shepherd for their thoughtful comments and feedback.

References

- [1] R. Kumar, S. Kishore, H. Lu, and A. Prakash, "Security analysis of unified payments interface and payment apps in india," in *29th USENIX Security Symposium (USENIX Security 20)*, 2020, pp. 1499–1516.
- [2] K. K. Lakshmi, H. Gupta, and J. Ranjan, "Upi based mobile banking applications—security analysis and enhancements," in *2019 Amity International Conference on Artificial Intelligence (AICAI)*, 2019.
- [3] NPCI, *National payments corporation of india official page*, <https://www.npci.org.in/what-we-do/upi/product-overview>, Accessed: 2024-01-10, 2024.
- [4] RBI, *Payments vision 2025*, <https://rbi.org.in/scripts/PublicationReportDetails.aspx?ID=1202>, Accessed: 2022-06-17, 2022.
- [5] S. Rastogi, A. Sharma, C. Panse, and V. M. Bhimavarapu, "Unified payment interface (upi): A digital innovation and its impact on financial inclusion and economic development," *Universal Journal of Accounting and Finance*, 2021.
- [6] PhonePe, *Phonepe application official page*, <https://www.phonepe.com/>, Accessed: 2024-02-15, 2024.
- [7] GooglePay, *Googlepay application official page*, <https://pay.google.com/about/>, Accessed: 2024-02-15, 2024.
- [8] UPINPCI, *Upi live members - list of banks live on upi_npci*, <https://www.npci.org.in/what-we-do/upi/live-members>, Accessed: 2024-01-20, 2023.
- [9] NPCI, *Upi transactions touch new record of 18.4-lakh crore in jan*, <https://www.thehindubusinessline.com/money-and-banking/upi-transactions-touch-new-record-of-184-lakh-crore-in-jan/article67804759.ece>, Accessed: 2024-03-15, 2024.
- [10] A. S. George, A. H. George, T. Baskar, and A. G. Martin, "An overview of india's unified payments interface (upi): Benefits, challenges, and opportunities," *Partners Universal International Research Journal*, 2023.
- [11] India.com, *Biggest scam ever! fraudsters steal rs 1 cr from 81 upi users in mumbai; know their modus operandi*, <https://www.india.com/maharashtra/biggest-upi-scam-in-mumbai-fraudsters-steal-rs-1-crore-from-81-upi-users-in-mumbai-kyc-scam-know-their-modus-operandi-5969266/>, Accessed: 2023-03-29, 2023.
- [12] TimesofIndia, *'money transferred by mistake': Beware, it's a trap of fraud*, <https://timesofindia.indiatimes.com/city/ludhiana/money-transferred-by-mistake-beware-its-a-trap-of-fraudsters/articleshow/98889538.cms>, Accessed: 2023-08-15, 2023.
- [13] HindustanTimes, *Received money on your upi app from a stranger? here's what you should do next*, <https://www.hindustantimes.com/technology/received-money-on-your-upi-app-from-a-stranger-here-s-what-you-should-do-next-101679137602393.html>, Accessed: 2023-08-10, 2023.
- [14] NPCI, *List of countries that accept upi internationally*, <https://www.npci.org.in/who-we-are/group-companies/npci-international/list-of-countries>, Accessed: 2024-01-15, 2023.
- [15] NPCI, *Upi is now accepted in france*, <https://www.npci.org.in/PDF/npci/press-releases/2024/Press-Release-UPI-is-Now-Accepted-in-France.pdf>, Accessed: 2024-01-20, 2023.
- [16] EPI, *Changing the way europe pays*, <https://www.epicompany.eu/>, Accessed: 2024-05-10, 2024.
- [17] Wero, *Move money in real-time for real*, <https://wero-wallet.eu/>, Accessed: 2024-05-10, 2024.
- [18] P. Ranjith, S. Kulkarni, and A. J. Varma, "A literature study of consumer perception towards digital payment mode in india," *Psychology and Education*, vol. 58, no. 1, pp. 3304–3319, 2021.
- [19] S. Khanra, R. P. Joseph, A. Dhir, and P. Kaur, "Antecedents of the barriers toward the adoption of unified payment interface," in *International working conference on transfer and diffusion of IT*, Springer, 2020, pp. 608–625.
- [20] Statista-FinancialServices, *Leading remitter banks for unified payments interface in india in june 2023, by volume(in millions)*, <https://www.statista.com/statistics/1406448/india-leading-upi-remitter-banks-by-volume/>, Accessed: 2023-06-01, 2023.
- [21] K. Neema and A. Neema, "Upi (unified payment interface)—a new technique of digital payment: An explorative study," *International Journal of Current Research in Multidisciplinary*, 2016.
- [22] Phonepe, *Phonepe application payment flow*, <https://developer.phonepe.com/v2/docs/payment-flow-1>, Accessed: 2023-12-10, 2023.
- [23] Statista-FinancialServices, *Unified payment interface (upi) usage across india in third quarter of 2023, by platform*, <https://www.statista.com/statistics/1034443/india-upi-usage-by-platform/>, Accessed: 2024-01-10, 2024.
- [24] D. M. Mahapatra and S. K. Patra, "Electronic payment adoption in india: Development and policy issues," *Aweshkar Research Journal*, 2018.
- [25] M. Kumar and J. Singh, "Digital banking in india-trend and challenges," *Think India Journal*, 2019.
- [26] S. Bhardwaj and M. Kaushik, "Unified payment interface—a way ahead for demonetization in india," in *Smart Computing and Informatics: Proceedings of the First International Conference on SCI 2016, Volume 2*, Springer, 2018, pp. 273–280.
- [27] R. B. Kakadel and N. A. Veshne, "Unified payment interface (upi)—a way towards cashless economy," *International Research Journal of Engineering and Technology*, 2017.
- [28] R. Gochhwal, "Unified payment interface—an advancement in payment systems," *American Journal of Industrial and Business Management*, 2017.
- [29] S. Chavan, P. Gore, and G. Bhutkar, "User survey of upi-enabled payment apps," in *International Conference of the Indian Society of Ergonomics*, Springer, 2021, pp. 1457–1469.
- [30] G. Sankararaman and S. Suresh, "A study on unified payment interface (upi) transactions (a digital banking tool) in chennai city," *Design Engineering*, pp. 10 431–10 440, 2021.

- [31] A. M. DAVID and M. E. A. M. LUSIA, "Users' preference towards g-pay app in madurai city," *Shodha Prabha (UGC CARE Journal)*, 2022.
- [32] B. Nagarjuna, "Emerging trends in digital payments—a prospective study," *ijemr.in*, 2019.
- [33] J. Thakkar and P. Thakkar, "Digital payments revolution: A study of awareness, acceptance, and usage of unified payments interface technology among selected women in india," in *2022 OPJU International Technology Conference on Emerging Technologies for Sustainable Development (OTCON)*, IEEE, 2023, pp. 1–6.
- [34] S. Vadlamudi and J. Sam, "Unified payments interface—preserving the data privacy of consumers," in *2022 International Conference on Cyber Resilience (ICCR)*, 2022.
- [35] P. Borborah and K. S. Das, "Building agency: Women vendors and gendered technology in informal markets in assam," *Gender & Development*, vol. 30, no. 3, pp. 517–530, 2022.
- [36] T. N. Jagatic, N. A. Johnson, M. Jakobsson, and F. Menczer, "Social phishing," *Communications of the ACM*, vol. 50, no. 10, pp. 94–100, 2007.
- [37] K. Sharma, N. Bajaj, X. Page, and M. Mondal, "A platform for uncovering indian users' decision-making process in unified payment interface (upi) apps," *Usenix.org*, 2022.
- [38] Y. Madwanna, M. Khadse, and B. Chandavarkar, "Security issues of unified payments interface and challenges: Case study," in *2021 2nd International Conference on Secure Cyber Computing and Communications (ICSCCC)*, 2021.
- [39] B. J. Fogg, J. Marshall, O. Laraki, *et al.*, "What makes web sites credible? a report on a large quantitative study," in *Proceedings of the SIGCHI conference on Human factors in computing systems*, 2001, pp. 61–68.
- [40] K. Browne, "Snowball sampling: Using social networks to research non-heterosexual women," *International journal of social research methodology*, vol. 8, no. 1, pp. 47–60, 2005.
- [41] PaytmApplication, *Paytm application official page*, <https://paytm.com/>, Accessed: 2024-02-15, 2024.
- [42] BharatInterfaceforMoney, *Bhim application official page*, <https://www.bhimupi.org.in/>, Accessed: 2024-02-15, 2024.
- [43] NPCI-BHIM, *Is there a limit to the amount of money that can be sent using bhim?* <https://www.npci.org.in/what-we-do/bhim/faq>, Accessed: 2024-01-15, 2024.
- [44] S. Elo and H. Kyngäs, "The qualitative content analysis process," *Journal of advanced nursing*, vol. 62, no. 1, pp. 107–115, 2008.
- [45] S. Takai and K. Ishii, "A use of subjective clustering to support affinity diagram results in customer needs analysis," *Concurrent Engineering*, vol. 18, no. 2, pp. 101–109, 2010.
- [46] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative research in psychology*, vol. 3, no. 2, pp. 77–101, 2006.
- [47] C. Auerbach and L. B. Silverstein, *Qualitative data: An introduction to coding and analysis*. NYU press, 2003, vol. 21.
- [48] U. Kuckartz, S. Rädiker, U. Kuckartz, and S. Rädiker, "Analyzing intercoder agreement," *Analyzing qualitative data with MAXQDA: Text, audio, and video*, pp. 267–282, 2019.
- [49] N. M. Bradburn, L. J. Rips, and S. K. Shevell, "Answering autobiographical questions: The impact of memory and inference on surveys," *Science*, vol. 236, no. 4798, pp. 157–161, 1987.
- [50] J. S. Olson and W. A. Kellogg, *Ways of Knowing in HCI*. Springer, 2014, vol. 2.
- [51] P. I. Fusch Ph D and L. R. Ness, "Are we there yet? data saturation in qualitative research," 2015.
- [52] B. Saunders, J. Sim, T. Kingstone, *et al.*, "Saturation in qualitative research: Exploring its conceptualization and operationalization," *Quality & quantity*, vol. 52, pp. 1893–1907, 2018.
- [53] Statista, *Insights and facts across 170 industries and 150+ countries*, <https://www.statista.com/>, Accessed: 2024-01-10, 2024.
- [54] ICICI, *Industrial credit and investment corporation of india, imobile official page*, <https://www.icicibank.com/mobile-banking/imobile-pay>, Accessed: 2024-02-15, 2024.
- [55] SBI, *State bank of india, yono official page*, <https://www.sbiyono.sbi/index.html>, Accessed: 2024-02-15, 2024.
- [56] Axis, *Axis bank, axis pay official page*, <https://www.axisbank.com/bank-smart/axispay-upi-app/bhim-axis-pay-upi-app>, Accessed: 2024-02-15, 2024.
- [57] DigitalIndiaCSP, *Customer service point official website*, <https://www.digitalindiacsp.in/>, Accessed: 2023-09-10, 2023.
- [58] GrahakSevaKendra, *Understanding customer service points*, <https://grahaksevakendra.in/customer-service-points-csps/#:~:text=Customer%20Service%20Points%2C%20often%20referred,extension%20of%20the%20bank's%20services.>, Accessed: 2023-09-10, 2021.
- [59] A. Kostan, S. Olschar, L. Simko, and Y. Acar, "Exploring digital security and privacy in relative poverty in germany through qualitative interviews," in *33rd USENIX Security Symposium (USENIX Security 24)*, 2024, pp. 2029–2046.
- [60] G. Pay, *Learn about smart routing with extra upi ids*, <https://support.google.com/pay/india/answer/10331134?hl=en>, Accessed: 2024-02-15, 2024.
- [61] PhonePe, <https://cms.phonepe.com/en/myhelp/bank-account-bhim-upi/bhim-upi/bhim-upi-id-vpa/why-do-i-need-activate-new-bhim-upi-ids/>, Accessed: 2024-02-15, 2024.
- [62] GooglePay, *Create a upi number on google pay*, <https://support.google.com/pay/india/answer/11428814?hl=en-GB#:~:text=A%20UPI%20Number%20is%20a,digit%20numeric%20ID%20you%20choose>, Accessed: 2024-12-10, 2024.
- [63] PhonePe, *How do i set my 8 or 9 digit upi number for my bank account on phonepe?* <https://cms.phonepe.com/en/myhelp/bank-accounts-bhim-upi-debitcredit-cards/bhim-upi-upi-number/setting-my-8-or-9-digit-upi-number/how-do-i-set-my-8-or-9-digit-upi-number-my-bank-account-phonepe/>, Accessed: 2024-02-15, 2024.
- [64] D. Bhattacharyya, R. Ranjan, F. Alisherov, M. Choi, *et al.*, "Biometric authentication: A review," *International Journal of u-and e-Service, Science and Technology*, vol. 2, no. 3, pp. 13–28, 2009.
- [65] E. M. Redmiles, A. R. Malone, and M. L. Mazurek, "I think they're trying to tell me something: Advice sources and selection for digital security," in *2016 IEEE Symposium on Security and Privacy (SP)*, IEEE, 2016, pp. 272–288.
- [66] L. Neil, H. S. Ramulu, Y. Acar, and B. Reaves, "Who comes up with this stuff? interviewing authors to understand how they produce security advice," in *Nineteenth Symposium on Usable Privacy and Security (SOUPS 2023)*, 2023, pp. 283–299.
- [67] E. M. Redmiles, S. Kross, and M. L. Mazurek, "How i learned to be secure: A census-representative survey of security advice sources and behavior," in *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*, 2016, pp. 666–677.
- [68] R. W. Reeder, I. Ion, and S. Consolvo, "152 simple steps to stay safe online: Security advice for non-tech-savvy users," *IEEE Security & Privacy*, vol. 15, no. 5, pp. 55–64, 2017.

- [69] D. Barrera, C. Bellman, and P. C. van Oorschot, "A close look at a systematic method for analyzing sets of security advice," *Journal of Cybersecurity*, vol. 9, no. 1, tyad013, 2023.
- [70] D. Barrera, C. Bellman, and P. Van Oorschot, "Security best practices: A critical analysis using iot as a case study," *ACM Trans. Priv. Secur.*, vol. 26, no. 2, 2023.
- [71] R. W. Reeder, I. Ion, and S. Consolvo, "152 simple steps to stay safe online: Security advice for non-tech-savvy users," *IEEE Security & Privacy*, vol. 15, no. 5, pp. 55–64, 2017.
- [72] J. Schmüser, H. Sri Ramulu, N. Wöhler, *et al.*, "Analyzing security and privacy advice during the 2022 russian invasion of ukraine on twitter," in *Proceedings of the CHI Conference on Human Factors in Computing Systems*, ser. CHI '24, Honolulu, HI, USA: Association for Computing Machinery, 2024.
- [73] J. Schmüser, H. Sri Ramulu, N. Wöhler, *et al.*, "Analyzing security and privacy advice during the 2022 russian invasion of ukraine on twitter," in *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2024, pp. 1–16.
- [74] Y. Acar, S. Fahl, and M. L. Mazurek, "You are not your developer, either: A research agenda for usable security and privacy research beyond end users," *2016 IEEE Cybersecurity Development (SecDev)*, pp. 3–8, 2016.
- [75] H. Mai and S. Schneider, "European autonomy in payments," 2023.
- [76] M. Finn and K. Shilton, "Ethics governance development: The case of the menlo report," *Social Studies of Science*, p. 03 063 127 231 151 708, 2023.
- [77] RBI, *Reserve bank of india official page*, <https://www.rbi.org.in/>, Accessed: 2024-02-15, 2024.
- [78] NPCI, *National payments corporation of india official page*, <https://www.npci.org.in/>, Accessed: 2024-02-15, 2024.
- [79] SBI, *State bank of india official page*, <https://sbi.co.in/>, Accessed: 2024-02-15, 2024.
- [80] ICICI, *Industrial credit and investment corporation of india official page*, <https://www.icicibank.com/>, Accessed: 2024-02-15, 2024.
- [81] HDFC, *Housing development finance corporation official page*, <https://www.hdfcbank.com/>, Accessed: 2024-02-15, 2024.
- [82] APGVB, *Andhra pragathi gramina vikas bank official page*, <https://www.apgvbank.in/>, Accessed: 2024-02-15, 2024.
- [83] Axis, *Axis bank official page*, <https://www.axisbank.com/>, Accessed: 2024-02-15, 2024.
- [84] BankofBaroda, *Bank of baroda official page*, <https://www.bankofbaroda.in/>, Accessed: 2024-02-15, 2024.
- [85] UnionBankofIndia, *Union bank of india official page*, <https://unionbankofindia.co.in/english/home.aspx>, Accessed: 2024-02-15, 2024.
- [86] PunjabNationalBank, *Punjab national bank official page*, <https://www.pnbindia.in/>, Accessed: 2024-02-15, 2024.
- [87] KotakMahindraBank, *Kotak mahindra bank official page*, <https://www.kotak.com/en/home.html>, Accessed: 2024-02-15, 2024.
- [88] CanaraBank, *Canara bank official page*, <https://canarabank.com/>, Accessed: 2024-02-15, 2024.
- [89] NPCI, *Npci upi safety shield advice*, [npci](https://npci.org.in/what-we-do/upi/upi-safety-shield), <https://npci.org.in/what-we-do/upi/upi-safety-shield>, Accessed: 2024-05-20, 2024.
- [90] NPCI, *Good practices for a safe digital banking experience*, [rbi.org.in](https://npci.org.in/what-we-do/upi/upi-safety-shield), <https://npci.org.in/what-we-do/upi/upi-safety-shield>, Accessed: 2023-12-10, 2023.
- [91] SBI, *Be aware and stay protected from cyber frauds. be an alert customer*, [sbi.co.in](https://www.sbi.co.in/web/personal-banking/cyber-security), <https://www.sbi.co.in/web/personal-banking/cyber-security>, Accessed: 2024-02-10, 2024.
- [92] SBI, *Dos and don'ts*, [sbi.co.in](https://www.sbi.co.in/documents/16012/14424788/220322-ENGLISH.pdf/ca64722c-d7c7-67c8-1279-e8006e156fd1?t=1647954480292), <https://www.sbi.co.in/documents/16012/14424788/220322-ENGLISH.pdf/ca64722c-d7c7-67c8-1279-e8006e156fd1?t=1647954480292>, Accessed: 2024-02-10, 2024.
- [93] ICICI, *Mobile banking safety tips & measures*, <https://www.icicibank.com/online-safe-banking/mobile-banking>, Accessed: 2024-05-10, 2024.
- [94] ICICI, *5 ways to improve the security of mobile banking apps*, <https://www.icicibank.com/blogs/mobile-banking/5-ways-to-improve-the-security-of-mobile-banking-apps>, Accessed: 2024-05-10, 2024.
- [95] HDFC, *A detailed guide to safe online banking*, <https://www.hdfcbank.com/personal/resources/learning-centre/secure/safe-online-banking-tips-measures>, Accessed: 2024-03-20, 2024.
- [96] HDFC, *Hdfc bank phonebanking advice*, <https://www.hdfcbank.com/personal/useful-links/security/security-tips/secure-phonebanking>, Accessed: 2024-03-20, 2024.
- [97] HDFC, *Password security tips. how to protect your banking password*, <https://www.hdfcbank.com/personal/useful-links/security/security-tips/password-security-tips>, Accessed: 2024-03-20, 2024.
- [98] HDFC, *Hdfc bank mobile security dos and don'ts*, <https://www.hdfcbank.com/personal/useful-links/security/security-tips/mobile-security>, Accessed: 2024-03-20, 2024.
- [99] HDFC, *Hdfc bank secure banking online advice*, <https://www.hdfcbank.com/personal/useful-links/security>, Accessed: 2024-03-20, 2024.
- [100] APGVB, *Apvgb cyber security awareness*, https://apgvb.in/cyber_awareness, Accessed: 2024-06-25, 2024.
- [101] Axis, *Tips to use upi safely and transact with ease*, <https://www.axisbank.com/progress-with-us/digital-banking/tips-to-use-upi-safely-and-transact-with-ease>, Accessed: 2024-07-05, 2024.
- [102] Axis, *Secure your online payments. keep your money safe*, <https://www.axisbank.com/progress-with-us/digital-banking/how-to-avoid-online-payment-fraud-while-using-upi-apps>, Accessed: 2024-07-05, 2024.
- [103] Axis, *Axis bank tips for safe banking practices*, <https://www.axisbank.com/progress-with-us/digital-banking/tips-for-safe-banking-practices>, Accessed: 2024-07-05, 2024.
- [104] Axis, *Protect yourself from upi fraud: Tips and strategies*, https://application.axisbank.co.in/WebForms/axis-support/Blogs/blog_3.aspx, Accessed: 2024-07-05, 2024.
- [105] BankofBaroda, *Bob world upi: Secure upi*, <https://www.bankofbaroda.in/personal-banking/digital-products/instant-banking/bob-world-upi>, Accessed: 2024-02-15, 2024.
- [106] BankofBaroda, *Bhim baroda pay advice*, <https://www.bankofbaroda.in/writereaddata/images/pdf/Dos-and-Dontsbaroda-m-pay.pdf>, Accessed: 2024-02-15, 2024.
- [107] BankofBaroda, *Baroda mpay advice*, <https://www.bankofbaroda.in/writereaddata/Images/pdf/upi-user-tips-for-pim-security.pdf>, Accessed: 2024-02-15, 2024.
- [108] UnionBankofIndia, *Union bank cyber security tips*, <https://www.unionbankofindia.co.in/english/cyber-security-tips.aspx>, Accessed: 2024-02-15, 2024.

- [109] P. ationalBank, *Punjab bank upi security tips*, <https://www.pnbindia.in/Unified-Payments-Interface.html>, Accessed: 2024-02-15, 2024.
- [110] PunjabNationalBank, *Fraudster alert. be a upi smart citizen*, <https://netpnb.com/>, Accessed: 2024-02-15, 2024.
- [111] KotakMahindraBank, *Here are some safety tips to ensure your upi payments are safe*, <https://www.kotak.com/en/safe-banking/upi-payments.html>, Accessed: 2024-02-15, 2024.
- [112] CanaraBank, *Canara bank upi frauds*, <https://canarabank.com/pages/upi-frauds>, Accessed: 2024-01-20, 2024.
- [113] CanaraBank, *Canara bank terms and conditions upi*, <https://canarabank.com/pages/term-&-conditions-upi>, Accessed: 2024-01-20, 2024.
- [114] PhonePe, *Phonepe your safety comes first*, <https://www.phonepe.com/trust-and-safety/>, Accessed: 2024-01-20, 2024.
- [115] PhonePe, *Phonepe phishing. medium*, <https://medium.com/phonepe/tagged/phishing>, Accessed: 2024-01-20, 2024.
- [116] GooglePay, *Protect your google pay information. support.google.com*, <https://support.google.com/pay/india/answer/7295786?hl=en&zipy=%2Cphishing%2Cspoofing%2Cfraudulent-money-requests>, Accessed: 2024-01-20, 2024.
- [117] GooglePay, *Protecting against fraud or unauthorised activity. support.google.com*, <https://support.google.com/pay/india/answer/9851787?hl=en>, Accessed: 2024-01-20, 2024.
- [118] Paytm, *Tips to use upi safely and transact with ease*, <https://paytm.com/blog/payments/upi/tips-to-use-upi-safely-and-transact-with-ease/>, Accessed: 2024-01-20, 2024.
- [119] Paytm, *Upi frauds: How to secure your account?* <https://paytm.com/blog/payments/upi/upi-frauds/>, Accessed: 2024-01-20, 2024.
- [120] Paytm, *Protect yourself from paytm account takeover fraud: Stay one step ahead of fraudsters*, <https://paytm.com/blog/fraud-safety/paytm-payment-bank-account-takeover-fraud/>, Accessed: 2024-01-20, 2024.
- [121] Paytm, *5 ways users can protect themselves from online fraud*, <https://paytm.com/blog/fraud-safety/users-can-protect-themselves-from-online-fraud/>, Accessed: 2024-01-20, 2024.
- [122] BHIM, *Upi safety shield tips by bhim. upichalega*, <https://upichalega.com/>, Accessed: 2024-01-20, 2024.
- [123] BHIM, *Bhim application advice*, <https://www.bhimupi.org.in/>, Accessed: 2024-01-20, 2024.
2. What are the most common types of transactions you make using UPI? [Probe - Any recurring payments set for bills (Gas, DTH, water, etc), booking flight tickets, trains, online shopping, food ordering, transactions with family and friends, etc.]
 3. Which UPI applications do you use? Is there any reason to use this application? [Probe - Stand-alone UPI apps or bank UPI apps]
 4. Which devices do you commonly use for UPI transactions? [Probe- Have you used anything other than your mobile phone?]
 5. If you use UPI regularly, walk us through your typical day and all the transactions you make using UPI. [Probe - Paying taxi fare for daily commute, food courts, etc.]
 6. How do you usually check if the UPI transactions are successful or not? [Probe - Both ways like receiving and sending money]

c) Information Sources and Learning

1. Do you recall your first experience with UPI? How did you learn to use it? [Probe - Did you receive or look for information on how to use UPI when you started using it?]
a) If yes, from where did you get that information?
2. What are some practices you follow to stay safe while doing UPI transactions? Where did you get this information from? [Example - Family/friends/colleagues, internet, advertisements]
3. Have you experienced any scams or frauds while using UPI, could you share your experiences?
a) If not, have you heard of any scams or frauds on UPI from others?
b) How did these experiences and stories impact you? Did you take any action to stay safe after hearing these stories? What extra care did you start taking after that?
4. What information do you wish you knew before using UPI?

d) Concerns and Challenges

1. Are there any challenges you face while using the UPI application? [Probe - with installation, updation, navigation, configuration]
2. Are there any specific scenarios where you are watchful (more careful) while making transactions? [Probe - paying to unknown/online merchant, paying using links sent via message or email]
a) What do you do to protect yourself? OR what do you do about it?
b) If they use more than one application, what unique challenges do they face with that application?
3. Are there any scenarios that you are concerned about while doing transactions but haven't done anything about it? [Probe - making transactions in a hurry, verifying the authentication of the receiver, etc.]
4. How do you handle any issues you come across while using UPI? (Example scenarios - a) money got debited from your account but doesn't credit to the receiver's account, b) sent money to the wrong person, etc.)
a) If so can you talk about a specific issue?
b) Are there any other scenarios you are concerned about? [Probe - Resolve using online information, ask friends/colleagues, contact customer support, approach law enforcement, etc.]
5. Did you receive any information or guidance from the UPI provider or bank while facing challenges?
6. Are there any particular challenges you experienced while making the transactions? [Probe - Internet issues, receiver device issues, QR code scanner issues etc.]

e) Safety Practices and Awareness

1. Can you give us the general safety steps you follow while using UPI? [Probe - Strong Pin, changing pin/password frequently, not sharing pin, updating UPI app with a new version, not clicking on any links, not proceeding with transactions unless validated]
a) From where did you learn these safety measures?
b) Which other measures do you wish you knew?

A Interview Guide

a) Participant's UPI Background

1. Can you tell us a bit about your background, about your education, work, etc.? [Note - In the consent form, demographics will be collected and include educational background details]
2. When and how did you first get to know about UPI?
3. Have you got any help signing up for the UPI app?
4. How long have you been using UPI? And why did you start using it?
5. How often do you use it?

b) UPI Usage Questions

1. What are the major reasons for using UPI? When do you use it most? [Probe - Business, Personal transactions]

2. Do you know any UPI settings to safeguard your transactions? If so, please name a few. [Probe - 2-factor authentication, biometric-enabled, screen-lock enabled]
 - a) From where did you learn these settings and configurations?
 - b) Which other settings or features do you wish you knew?
3. Have you ever lost your phone while having UPI applications installed in it?
 - a) If Yes, then what concerns did you have? And what measures did you take?
 - b) If No, do you know what steps to take if you lose?
4. Have you ever given your phone to repair/exchange while having UPI applications installed in it?
 - a) If Yes, then what concerns did you have? And what measures did you take?
 - b) If No, do you know what steps to take if you give the phone for repair?
5. In your opinion, what measures would have helped you to develop awareness about UPI safety?
6. Is there anything else about UPI that we should have asked about or that you want to tell us?

B Advice Analysis

Table 2: Banks and UPI providers used for Official Advice Analysis.

Entities	Entity Names	Governing Body
UPI Regulatory Bodies	a) Reserve Bank of India (RBI) [77]	Public Entity
	b) National Payments Corporation of India (NPCI) [78]	Public Entity
Banks with UPI	a) State Bank of India (SBI) [79]	Public Entity
	b) Industrial Credit and Investment Corporation of India (ICICI) [80]	Private Entity
	c) Housing Development Finance Corporation (HDFC) [81]	Private Entity
	d) Andhra Pragathi Gramina Vikas Bank (APGVB) [82]	Public Entity
	e) Axis Bank [83]	Private Entity
	f) Bank of Baroda [84]	Public Entity
	g) Union Bank of India [85]	Public Entity
	h) Punjab National Bank [86]	Public Entity
	i) Kotak Mahindra Bank [87]	Private Entity
	j) Canara Bank [88]	Public Entity
UPI Applications	a) PhonePe [6]	Private Entity
	b) Google Pay [7]	Private Entity
	c) Paytm [41]	Private Entity
	d) Bharat Interface for Money (BHIM) [42]	Public Entity

Table 3: Official Advice from Banks and UPI Providers.

Entities	Scraped Advice
NPCI	UPI Safety Shield [89]
RBI	Good practices for a safe digital banking experience [90]
SBI	a) Steps taken by SBI to Ensure your online security [91]
	b) Your role in ensuring your digital security [91]
	c) Phishing [91]
	d) Vishing [91]
	e) Smishing [91]
	f) Mobile Security [91]
	g) Do's and Don'ts [92]
ICICI	a) Mobile Banking Safety Tips and Measures [93]
	b) 5 ways to improve the security of Mobile Banking apps [94]
HDFC	a) A Detailed Guide To Safe Online Banking [95]
	b) Phonebanking [96]
	c) Password Security Tips [97]
	d) Mobile Security – Dos And Don'ts [98]
	e) Secure Banking Online [99]
APGVB	Cyber security awareness [100]
Axis Bank	a) Tips to use UPI safely and transact with ease [101]
	b) Secure your online payments. Keep your money safe [102]
	c) Tips for safe banking practices [103]
	d) Protect yourself from UPI fraud [104]
Bank of Baroda	a) Secure UPI [105]
	b) Do's and Don'ts [106]
	c) Security PINs, Password security guidelines [107]
Union Bank of India	Cyber security tips [108]
Punjab National Bank	a) Security tips [109]
	b) Do's and Don'ts [110]
Kotak Mahindra Bank	Safety Tips for UPI payments [111]
Canara Bank	a) UPI frauds [112]
	b) Term and Conditions UPI [113]
PhonePe	a) Your safety comes first [114]
	b) Phishing [115]
Google Pay	a) Protect your Google Pay information [116]
	b) Protecting against Fraud or Unauthorized activity [117]
Paytm	a) Tips to use UPI safely and transact with ease [118]
	b) UPI Frauds: How to Secure Your Account? [119]
	c) Stay One Step Ahead of Fraudsters [120]
	d) 5 Ways Users Can Protect Themselves From Online Fraud [121]
BHIM	a) UPI Safety Shield [122]
	b) Do's and Don'ts [123]

C Interview Codebook

Table 4: Interview Codebook.

Category	Codes	Definition or example	Category	Codes	Definition or example
Participant Background	Work background	Profession of participants	Guidance & Awareness	Handling UPI issues	Contact official help desk, discuss with family, friends, report suspicion
	Education background	Education of participants		Guidance from UPI provider/banks	Don't share sensitive details with officials, follow guidelines, avoid unknown calls, emails, texts, links
	UPI usage frequency	No: of transactions in a day		Users preferred awareness platforms	Official websites, social media, official contacts from banks, UPI providers
	UPI usage duration	No: of months/years of usage	Concerns & Challenges	Measures to increase awareness	Increase campaigns, prioritize targeted promotions, educate over social media, other sources
	Mobile OS	Adroid or iOS		Miscellaneous statements	Few supporting statements made by participants
Initial Signup &Source	Devices	Devicetypes- mobile, tab or laptop/PC		Concerns	Risks with PIN-less transactions, the authenticity of QR codes, payment links, etc
	UPI apps	Apps they used so far	Challenges	Challenges	Server glitches, internet issues, scanner problems, transaction failures, cybercrime cases
	Knowledge gap	Participant knowledge gap with settings and features		Watchful scenarios	Paying with links, QR codes, high volume transactions, unknown merchants
Usage Motivations	Reason of UPI usage	Like government mandates, pandemic, cashless movements etc	Scams & Frauds Details	Concerns with UPI settings	Avoid auto-pay options, forgetting setting changes, etc
	Source of Info	Family, friends, ads, promos, etc		Technical challenges	Issues with QR, lack of KYC, server problems, etc
	Source of learning	Google, friends, family, documents, self-learned, etc		App-specific challenges	Bank server downtimes, ease of use, lot of options, etc
Transaction Types	Reason for using	Rewards, cashback, ease of use	Experienced scams/frauds with UPI	Heard/read experiences	Sharing OTP on call, transactions with links, other experiences
	Usage reasons of UPI	Fast, secure, cashless, easy, avoid banks, ATMs, cash		Personal experiences	Its Yes or No, whether participants experienced scams/frauds
	Day-to-day transactions	Food, online shopping, transportation, sending to family, friends		Major source of scams	Clicking on unknown links, posting on the online marketplace, job offers, cashback, etc
Transaction Verification	Recurring/auto-debit	Entertainment subscriptions, maintaining a minimum balance, bill reminders	Impact of scams, frauds		Online marketplace, calls, advertisements, sharing OTP, PIN, etc
	Types of transactions	Ticket bookings, online shopping, food ordering, paying bills, sending to family, friends			Ignore unknown calls, links, block suspicious contacts, don't share OTP, PIN, QR codes, verify receiver details
	Receiving- success verification	Check the history, account balance, confirmation from the receiver, the notification from banks, UPI			Uninstall banking apps, hiding apps, removing SIM cards, taking backups, switching off mobile, removing battery, etc.
Transaction Verification	Sending- success verification	Check account balance, history, notifications from banks, UPI, confirmation from receiver	Handling Lost/Repair Devices	Safety measures before mobile repair	Repair technicians knowledge for hacking device etc
	Sending- failure verification	Check history, notifications		Safety measures after losing phone	Block account, changing PIN online, blocking all apps, lodging police complaint, etc
	Source of settings information	Google search, FAQs, social media, official guidance from UPI, self-learned		Concerns after losing phone	Trusting reboot process, confidence on locks, losing contacts
Safety Measures Sources	UPI settings for safety	2-factor, biometric authentications, PIN safety	Lost mobile	Given mobile for repair	Whether participant lost mobile or not, its Yes or No
	Source of UPI safety measures	Own intuition, social media, friends, family, information from banks, UPI providers			Whether participant gave mobile for repair or not, its Yes or No
	Security measures by users	Avoid sharing sensitive details- OTP, PIN, careful in public, double-check receiver details, etc			
Rationale	Rationale provided	Whether participants explained their choices of using or not using security settings, explanations for concerns, and advice they receive			
	No Rationale	When participants mentioned a security choice they made without providing further explanation.			